

1st IEEE VR workshop on Archaeology and Cultural Heritage through XR ARCHERIX

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ABSTRACT

The goal of this interdisciplinary workshop is to bring together in one place researchers specialized in XR and Cultural Heritage users such as archaeologists, restorers and curators, who until now have often expressed themselves collectively but separately through a number of scientific works. As each participant will contribute his or her own expertise, experience and point of view, the workshop should serve to strengthen cooperation and partnerships between disciplines.

1 ORGANIZERS

Selma Rizvić is a Professor of Computer Graphics at the Faculty of Electrical Engineering, University of Sarajevo. She has been teaching there Bachelor, Master, and Ph.D. study programs. In 2004 she founded the Laboratory for Computer Graphics - Sarajevo Graphics Group (SGG), a research group that specializes in the use of IT technologies for the presentation of cultural heritage. The SGG research group is multidisciplinary, it includes computer graphics experts and works with archaeologists, historians, visual artists, writers, and film professionals, in order to design and implement virtual cultural heritage applications that have educational values and are engaging, entertaining, and attractive at the same time. In 2010 they founded the Association for Cultural Heritage Digitization DIGI.BA.

Valérie Gouranton is a senior lecturer since 1999 and her research is attached to IRISA (Institut de Recherche en Informatique et Systèmes Aléatoires, UMR 6074) and Inria at the University of Rennes, while her teaching is attached to INSA (Institut National des Sciences Appliquées) Rennes. Her research focuses on extended reality (XR), and more specifically on models and tools for producing XR applications. Her main areas of interest are health, cultural heritage and the digital arts, but she is also interested in applications in construction and sport. She is in charge of the Arts, Culture and Heritage cross-disciplinary area at IRISA, the local coordinator for INSA Rennes of DemoES AIR (Augmenting Interactions in Rennes) and a member of its executive committee. She is a member of the executive committee of EquipEx+ Continuum, a collaborative research infrastructure of 30 platforms located throughout France, designed to advance interdisciplinary research between

computer science and the humanities and social sciences. She is in several Program Committees, such IEEE VR.

Holger Graf is head of the VR and AR department at the Fraunhofer Institute for Computer Graphics Research, Darmstadt Germany. He has a longstanding record in national and international EU-funded projects especially in the digital cultural heritage domain in which he was responsible for the administrative and scientific coordination of research on XR applied to Cultural Heritage, defining strategic agendas for the digitisation and the deployment of XR to the CH domain and applied XR solutions within the whole metaverse of the CH domain. He is co-organiser of several scientific events of applied computer graphics to cultural heritage such as the Eurographics (EG) Workshop on Computer Graphics for Cultural Heritage (GCH) series 2014, 2023, 2024 as well as the IEEE/EG World Congress on Digital Heritage in 2013, 2015 and 2025.

Théophane Nicolas is an archaeologist, ceramologist specialist in the Bronze Age of Western Europe at French national Institute for Preventive Archaeological Research (Inrap), in Trajectoires research laboratory, France. His research focuses on developing new approaches and tools for archaeology based on medical imaging (CT-scan) and Extended Realities.

Ronan Gagne is a research engineer at University of Rennes since 2010, and his research is attached to IRISA (Institut de Recherche en Informatique et Systèmes Aléatoires, UMR 6074) and Inria at the University of RennesFrance. He received a PhD degree in computer science from Rennes University, France, in 1997. He is the technical director of the Immersia research platform in virtual reality, member of the French National Research Infrastructure Continuum. His research interests are applications of XR to art, culture and heritage.

2 WORKSHOP MANIFESTO

In the same way as 3D, innovative uses for XR are beginning to emerge in the field of historical sciences, on the initiative of archaeologists, anthropological biologists, paleopathologists and conservator-restorers. These technologies, which are constantly progressing, are opening up a new era in the knowledge, conservation and dissemination of archaeological and more generally cultural heritage data. Fundamentally interdisciplinary, they are revolutionizing practices and raising new concerns and issues. XR based methods provide powerful interactive and non-destructive access to fragile historical material and immediate information. Furthermore, new approaches that combine AI methods to XR might benefit the cultural heritage domain, providing high visual quality solutions that allow for digital restoration, prediction of future (de)generative natural processes, scientific exploration and communication to a broader audience. In addition, XR provides a powerful opportunity for the preservation and dissemination of intangible heritage. With the increasing development of virtual reconstruc-

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tions in the domain of Cultural Heritage, and thanks to advances in the production and 3D animation of virtual humans, immersive and interactive simulations have emerged to preserve human activity associated with traditional culture and folklore.

However, the production of heritage related environments in XR raises a number of questions of both a methodological and a technical nature. Indeed, the nature and construction of such interactive 3D digital environments combine real elements, or at least those obtained by digitizing the real thing, with interpreted elements. The latter may be linked to a previous complete environment to represent missing elements, or to data created by the user during virtual work sessions to enrich the environment with the help of tools offered to him. It is important to define a methodology for controlling and representing these different levels of data in the XR environment, and to assess the cognitive relationship between the expert user and these representations.

3 PROGRAM OVERVIEW

The ARCHERIX workshop is structured around two high-quality keynote presentations, two sessions of selected presentations, one titled "Methodology and innovative approach", and one titled "Case study", and one panel session.

The first keynote presentation, given by Sofia Pescarin, is entitled "Universal concepts as triggers in the design of VR/MR applications for cultural heritage", and focuses on the challenges and perspective of designing interactive and immersive applications for cultural heritage based on cognitive and emotional goals. **Sofia Pescarin**, PhD in History and Computing, is currently a Senior Researcher at CNR ISPC. Her research interests cover a wide domain that includes Virtual Archaeology, Digital Heritage, Virtual Museums, Embodied Museology, Applied Games, Interactive Media Design. She is Chief Editor since 2017 of the Elsevier Journal "Digital Application in Archaeology and Cultural Heritage" (DAACH). Since 2018, she is a professor at the University of Bologna, where she teaches Intangible Heritage and Multimedia. She is currently coordinating the european project PERCEIVE, WP3 of the PNRR project CHANGES / Spoke 4 (Virtual Technologies) and the Task prototyping platforms for virtual exhibitions within the project PNRR H2IOSC.

The second keynote presentation, given by Maurizio Forte, is entitled "Neurophysiology and human perception: VR-AI experiments in archaeology". **Maurizio Forte**, PhD, is William and Sue Gross Distinguished Professor of Classical Studies Art, Art History, and Visual Studies at Duke University. He was a pioneer in digital archaeology and the first archaeologist working in a Supercomputing Center (CINECA) in the 80-90s. He was recently awarded the Saint Francis Prize in Techno-Humanities 2024 for his work in the field of Digital Humanities. This Prize is to be presented to individuals or institutions in recognition of their outstanding contributions to the development of technologies in the humanities for the well-being of humankind.

The session "Methodology and innovative approach" gathers four papers, one research paper, one technical paper, and two position papers. These papers were selected for their exploration of heritage research that include XR and the paradigm shift that 3D brings to archaeologists: digitisation, hypothesis formulation, knowledge organization and information visualization, and interaction tools.

The session "Case study" gathers four papers, two research papers, one technical paper, and one "work-in-progress paper. They illustrate widely differing aspects of the integration of XR into research, dissemination, and mediation in archaeology and cultural heritage.

The closing panel session aims to create a valuable cross-community discussion about the opportunities for collaboration between XR and the archaeology and cultural heritage domains, the issues and challenges of this interdisciplinary collaboration, and the

potential for innovation brought by XR in relation to AI for archaeology and cultural heritage.

4 PROGRAM COMMITTEE

An international program committee, made up of nine experts in the fields of XR, archaeology, and digital heritage, was responsible for reviewing the papers submitted:

- Maria Roussou, University of Athens, Greece
- Karina Rodriguez Echavarria, University of Brighton, United Kingdom
- Mai Otsuki, AIST, Japan
- Gabriele Guidi, Indiana University, USA
- François Giligny, Université Paris 1-Panthéon Sorbonne, France
- Bernd Froehlich, Bauhaus-Universität Weimar, Germany
- Nicolo Dell'Unto, Lund University, Sweden
- Marc Macé, CNRS, France
- Jean-Baptiste Barreau, CNRS, France

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